MTH357

Timed Online Assignment

(January 2020 Semester- Answer Keys):
For students' practice and reference

Question 1 (Total: 25 marks)

In order to effectively manage the advertising expense and make revenue projections, the manager of a company which sells children clothes online, collected data from 35 sales periods. The revenue (in million SGD) for the company (y) and advertisement costs (in million SGD) (x) are recorded for each sales period. You may assume data from all months are independent.

Using this data, the following regression model is estimated:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \ , \ \varepsilon_i \sim N(0, \sigma^2).$$

Here are some of the reported summary statistics:

$$\sum_i x_i y_i = 637.2153$$
$$\sum_i x_i^2 = 173.0992$$
$$\sum_i y_i^2 = 2743.15$$
$$\bar{x} = 2.115285$$
$$\bar{y} = 7.991667$$

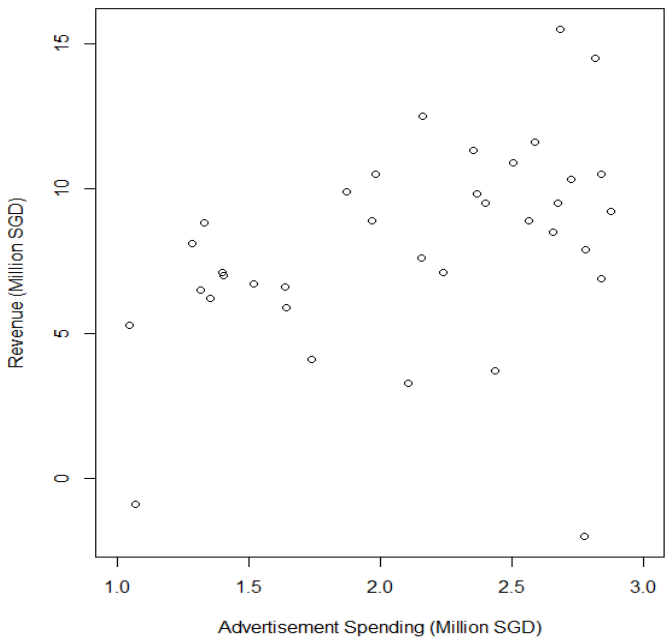


Figure Q1

(a) Estimate the coefficients β_0 and β_1 and interpret the point estimates.

(5 marks)

(b) Given that the estimated standard errors of β_0 and β_1 are 2.1024 and 0.9588 respectively, test the significance of the coefficients at $\alpha = 0.05$. Show all your steps.

(8 marks)

(c) State the distributional assumptions, if any, for the analysis in Question 1(b).

(2 marks)

(d) Suppose that there is a difference of 1.5 million SGD spent for advertisement in two randomly selected months.

(i) What is the expected difference in the revenue?

(2 marks)

(ii) Construct a 95 % CI for the difference in their expected response.

(8 marks)

Marking Guide:

(a)

$$S_{xy} = \sum xy - \frac{\sum x \sum y}{n} = 28.64775$$
$$S_{xx} = \sum x^2 - \frac{\sum x \sum x}{n} = 12.0197$$
$$\widehat{\beta_1} = \frac{S_{xy}}{S_{xx}} = 2.3834$$
$$\widehat{\beta_0} = \text{mean(y)} - \widehat{\beta_1} * mean(x) = 2.950096$$

$\widehat{\beta_0}$ is the expected revenue when there is no advertisement spending, however, it is extrapolated because advertisement spending is never 0. $\widehat{\beta_1}$ is the expected increase in revenue when for every 1 million increase in advertisement spending.

(b)

1. Write down the hypotheses.

$$H_0: \beta_j = 0 \; vs \; H_1: \beta_j \neq 0 \;\; \text{for } j = 0 \; and \; 1$$

2. Calculate the test statistic.(t-values in R output below)

3. Determine the critical region/confidence interval/p-value.
p-values in output below,
critical region: t=2.032245 (2 tailed).
CI: beta0: -1.322 to 7.22
CI: beta1: 0.434 to 4.3319

```
> summary(q1)

Call:
lm(formula = y ~ x)

Residuals:
     Min       1Q   Median       3Q      Max 
-11.5669  -0.8287   0.2893   2.0018   6.1457 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)   2.9501     2.1024   1.403    0.170  
x             2.3834     0.9588   2.486    0.018 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.324 on 34 degrees of freedom
Multiple R-squared:  0.1538,	Adjusted R-squared:  0.1289 
F-statistic:  6.18 on 1 and 34 DF,  p-value: 0.018
```

4. Write down a conclusion.
   At alpha = 5 %, the beta1 coefficient is statistically significant because p-value is lower than 5 %. Thus we can conclude that increasing the advertisement spending is associated with increased revenue, at least within the range of 1m to 3m advertisement spending.

(c) Assumption: The estimated betas are of normal distribution.

(d)

(i) Expected difference = 2.3834*1.5 = 3.5751 million SGD.

(ii) Let the revenue of the 1st month be Y_1 and the other month be Y_2

$$Var(Y_1 - Y_2) = Var(1.5b_1) = 2.25\, Var(b_1)$$
$$= 0.9588^2 * 2.25 = 2.068519$$

$$SE_{Y_1 - Y_2} = sqrt(2.068519) = 1.4382$$

95% CI is

$$3.5751 \pm t_{0.975,34} * SE_{Y_1 - Y_2}$$
$$= 3.5751 \pm 2.0322 * 1.4382$$
$$= (0.65239, 6.49781) \text{ million SGD}$$

Question 2 (Total: 25 marks)

Accidents in aeroplanes are very rare but can lead to many fatalities in each incident. An analyst wishes to apply linear regression to analyze how the number of aeroplace incidents have changed from the period of 1988-1999 to the period of 2000 to 2014.

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \text{ , } \varepsilon_i \sim N(0, \sigma^2)$$

Where $y_i = incidence\ rate\ for\ airline\ i\ in\ 2000\ to\ 2014$
$x_i = incidence\ rate\ for\ airline\ i\ in\ 1988\ to\ 1999$

The sample size is 56, presenting 56 different airlines spanning across the globe. Also note that we are using incidence rate instead of number of incidents, because some airlines fly a lot more and thus should have higher number of incidents even with similar incidence rate.

The estimated equation is

$$\underset{(0.72297)}{E(y) = 2.88935} \quad + \quad \underset{(0.05324)}{0.15543\, x}$$

$R^2 = 0.1363$, $\bar{x} = 7.915, \bar{y} = 4.119$, SSE = 1043.755

However, the analyst feels that the simple linear regression model is not good enough. Thus, she proposes to include a new factor:

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - 10) z$$

where

$$z = \begin{cases} 1, & \text{if x} > 10 \\ 0, & \text{else} \end{cases} .$$

With the addition of this new factor, the regression becomes a piecewise linear regression model. The estimates (standard error) for β_0, β_1 & β_2 are 1.2327(1.047), 0.538 (0.187) & -0.484(0.2275) respectively.

(a) Write down the models for cases $x > 10$ and $x \leq 10$.

(4 marks)

(b) Sketch E(y) against x. x should lie on the horizontal axis. Label all key points of the plot.

(3 marks)

(c) Given that the SSE of this new model is 961.53, perform a nested F-test and determine if the new covariate is worth adding into the model.

(5 marks)

The analyst wishes to improve the model further by adding in one more covariate. The covariate "Asia" is an indicator variable tagged as 1 if the airline is headquartered in Asia, and 0 if not.

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - 10)z + \beta_3 Asia$$

```
Residuals:
    Min      1Q  Median      3Q     Max 
-6.3030 -2.2161 -0.0762  1.4523 20.0225 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept)   0.6565     1.0730   0.612  0.54331   
x             0.5016     0.1842   2.723  0.00879 **
(x-10)x1     -0.4366     0.2243  -1.946  0.05703 . 
Asian         2.1525     1.1852   1.816  0.07511 . 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.17 on 52 degrees of freedom
Multiple R-squared:  0.2518,	Adjusted R-squared:  0.2087 
F-statistic: 5.834 on 3 and 52 DF,  p-value: 0.001633
```

(d) Using the information from the R-output of this new model in Question 2(c), sketch this new model into the sketch drawn in part b. Label it clearly so that we can clearly see the differences between this model and the previous. Interpret the findings.

(8 marks)

(e) Instead of using a dichotomous variable "Asian", the analyst decides to have a new country variable. In the dataset, there are 56 airlines from 23 countries. Discuss whether this approach is feasible. If not, suggest a better way to include more country data into the model.

(5 marks)

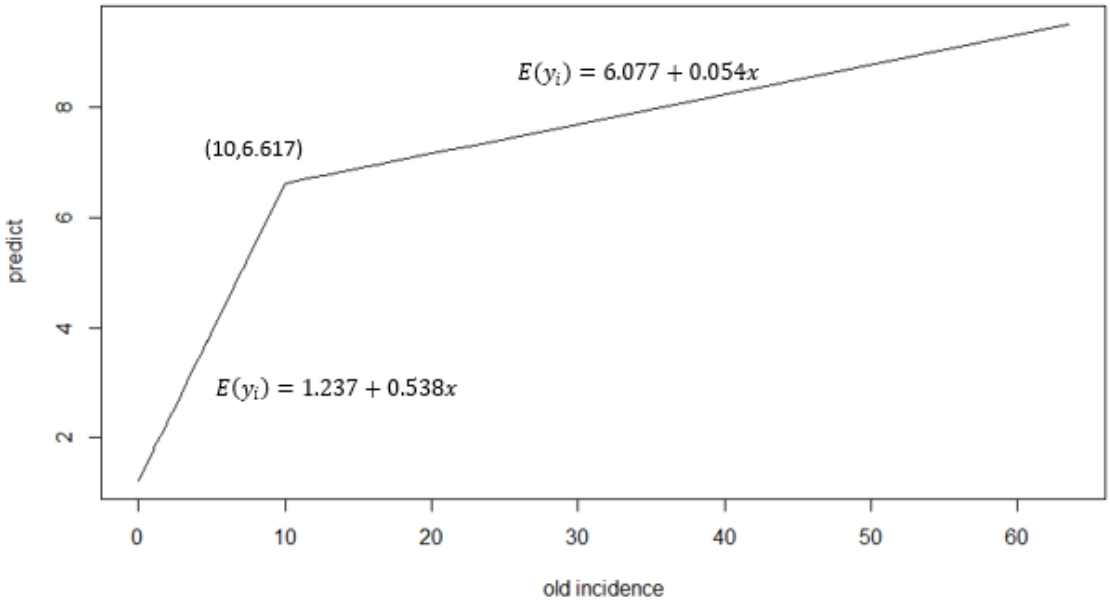
<u>Marking Guide:</u>

(a)

$$E(y) = 1.237 + 0.538x - 0.484(x - 10)z$$

If x > 10, $E(y) = 1.237 + 0.538x - 0.484(x - 10) = 6.077 + 0.054x$
If x ≤ 10, $E(y) = 1.237 + 0.538x$

(b) Sketch E(y) against x. x should lie on the horizontal axis. Label all key points of the plot.



(c) Given that the SSE of this new model is 961.53, perform a nested F-test and determine if the new covariate is worth adding into the model.

$$H_0: \beta_2 = 0 \; vs \; H_1: \beta_2 \neq 0$$

$$F = \frac{\frac{1043.755 - 961.53}{3-2}}{\frac{961.53}{56-3-1}} = \frac{82.22}{18.49} = 4.4467 \; (p - val = 0.0398)$$

Critical region 5% = $F_{0.975}(1,52)$ = 4.026631

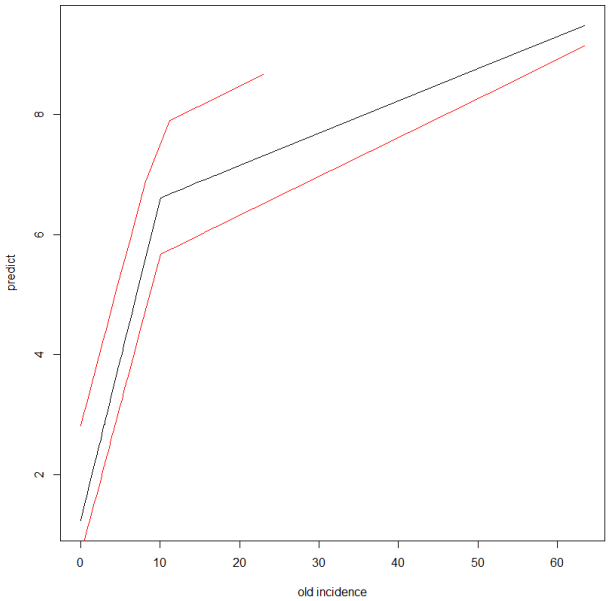
We reject the null hypothesis and conclude that the additional variable is worth adding into the model.

(d) Calculate the equations for Asian =1 and Asian = 0, x>10 and x<10.

$$E(y_i) = 0.6565 + 0.5016x - 0.4366(x - 10)z + 2.1525Asia$$

Asian = 1
If x > 10, $E(y) = 7.7555 + 0.065x$
If x ≤ 10, $E(y) = 2.809 + 0.5016x$

Asian = 0
If x > 10, $E(y) = 5.603 + 0.065x$
If x ≤ 10, $E(y) = 0.6565 + 0.5016x$



Compared to the previous model, we have separated Asian airlines and non-Asian airlines. The 2 lines differ by a translation of 2.1225 units along the Y axis. This implies that Asian airlines are in general, more incident prone. However, as non-Asian and Asian gradients are the same, it means that the effect of time (from 88-99 to 00-14) is the same for both.

(e) Using all 23 countries as 23 factors is unlikely to succeed as there are insufficient data with only 56 points. A better way is to group the countries by perhaps continents, or we may rank the countries in other of "1st world", "2nd world" etc. This will better make use of the data without using too many factors.

Question 3 (Total: 25 marks)

In this question, we will handle some theoretical concepts in regression analysis.

(a) Consider this simple linear regression mode:

$$y_i = \beta_1(8 + x_i) + \varepsilon_i \text{ , } \varepsilon_i \sim N(0, \sigma^2), \sigma\{\varepsilon_i, \varepsilon_j\} = 0 \; for \; i \neq j.$$

Using calculus, prove that the ordinary least squares estimator for β_1 is

$$\hat{\beta}_1 = \frac{\sum_i y_i(8 + x_i)}{\sum_i (8 + x_i)^2}$$

(10 marks)

(b) In the big data era, analysts are handling increasingly large datasets – both in the sample size and the number of covariates.

(i) Can the variance of the mean response be reduced to 0 if the sample size is large enough? (This is the variance used for finding confidence intervals)

(5 marks)

(ii) Can this variance be reduced to 0 with a large sample size? (This is the variance used for finding prediction intervals)

(3 marks)

(iii) Discuss the implications of Question 3(a) and 3(b).

(3 marks)

(c) Consider the multiple regression model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i.$$

Usually, when we use R to perform a regression analysis, we will use the lm-function to find all 4 betas.

In the situation where we already know that $\beta_2 = 20$, suggest how we can continue to use R to perform this analysis using standard code (such as the lm function).

(4 marks)

<u>Marking Guide:</u>

(a) Sum of squared error: $Q = \sum \left(y_i - \hat{\beta}_1(8 + x_i)\right)^2$

$$\frac{dQ}{d\hat{\beta}_1} = 2\sum \left(y_i - \hat{\beta}_1(8 + x_i)\right)(-1)(8 + x_i) = 0$$

$$\sum y_i(8 + x_i) = \sum \hat{\beta}_1(8 + x_i)^2$$

$$\hat{\beta}_1 = \frac{\sum y_i(8 + x_i)}{\sum(8 + x_i)^2}$$

(b) In the big data era, analysts are handling increasingly large datasets – both in the sample size and the number of covariates.

(i) Can the variance of the mean response, $Var(\hat{\mu}_{Y|X_h})$ be reduced to 0 if the sample size is large enough? (This is the variance used for finding confidence intervals)

$$Var(\hat{\mu}_{Y|X_h}) = \sigma^2\left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{\text{XX}}}\right)$$

As $n \to \infty, \frac{1}{n} \to 0, S_{XX} \to \infty$ (because S_{XX} is a sum)

$$\lim_{n\to\infty} Var(\hat{\mu}_{Y|X_h}) \to 0$$

Yes the variance of the mean response can be reduced to 0.

(ii) How about the Variance of the predicted Y_{new}? Can this variance be reduced to 0 with a large sample size? (This is the variance used for finding prediction intervals)

$$Var(Y_{new}) = \sigma^2\left(1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{\text{XX}}}\right)$$

As $n \to \infty, \frac{1}{n} \to 0, S_{XX} \to \infty$ (because S_{XX} is a sum)

$$\lim_{n\to\infty} Var(Y_{new}) \to \sigma^2$$

No, the variance of the predicted response cannot be reduced to 0.

(iii) Discuss the implications of a and b.

1. The predicted equation will become a systematic relationship linking Y and X. And the estimated beta-hats will be equal to the true betas.
2. There will always be uncertainty in estimating a new estimate because Y and X has random errors. Having a large sample size does not reduce the random errors in the population.

(c) If $\beta_2 = 20, 20x_{2i}\ becomes\ a\ constant$

$$y_i = \beta_0 + \beta_1 x_{1i} + 20x_{2i} + \beta_3 x_{3i} + \varepsilon_i$$

We can then create a new outcome variable, $y_i' = y_i - 20x_{2i}$, and continue to do the regression $y_i' = \beta_0 + \beta_1 x_{1i} + \beta_3 x_{3i} + \varepsilon_i$ in R.

Question 4 (Total: 25 marks)

An analyst is looking at a dataset which relates Wine scores with the amount of substances in the Wine. She wishes to perform diagnostics for leverage and influence on this data, and is asking you to help her with it.

y: Wine scores (from 1 to 10)
x_1: Citric Acid (sometimes added to wines to increase acidity)
x_2: Sulfites (antioxidant and antibacterial)
x_3: Alcoholic content

The regression model used is: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i$

The slope coefficients are all positive and statistically significant, with $\hat{\beta}_2 > \hat{\beta}_3 > \hat{\beta}_1$. The intercept coefficient is negative.

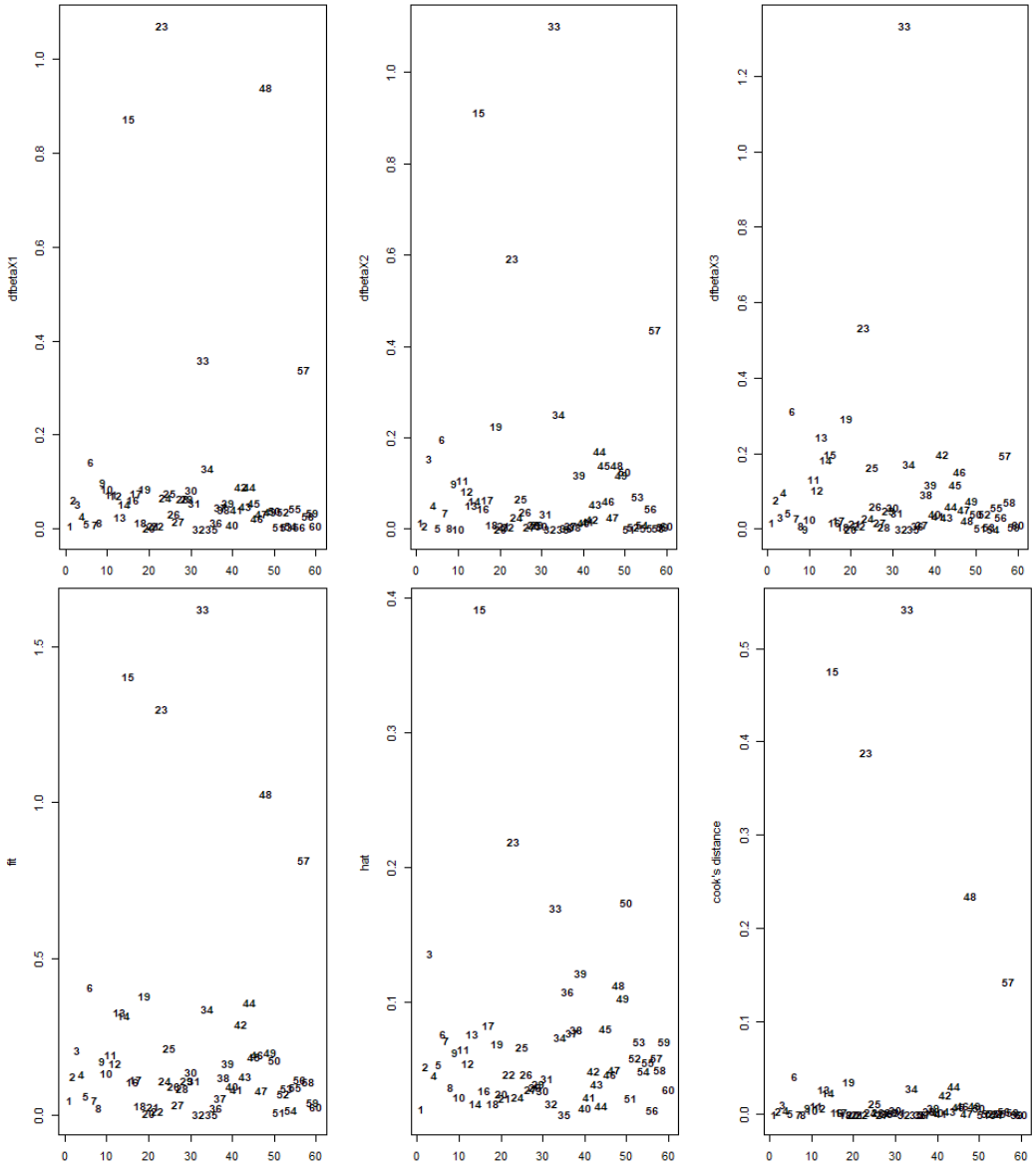


Figure Q4

The output from influence.measures() in R is plotted and labelled with the sample number for your convenience. The top three are the DFBETAS for Citric Acid, Sulfites and Alcohol respectively. The bottom three are the DFFITS, diagonals of the Hat matrix and Cook's Distance respectively.

(a) Explain what is influence and what does an influential point do to your model?

(4 marks)

(b) Perform an analysis of influence on the individual coefficients, showing your working and justification clearly. Identify potential outliers.

(6 marks)

(c) Perform an analysis of influence on the model as a whole, showing your working and justification clearly. Identify potential outliers.

(6 marks)

(d) From your analysis in Question 4(b) and 4(c), you will have identified potential outliers. Also attached in this question is the raw dataset which was used by the analyst. By referring to the raw dataset, discuss on the outliers that you have observed.

(6 marks)

(e) What remedial measures do you recommend?

(3 marks)

n	Score	Acid	Sulf	Alco	n	Score	Acid	Sulf	Alco
1	4.5	0.1683	0.6328	12.7510	31	2.2	0.0264	0.4586	11.1107
2	2.2	0.0000	0.5932	10.7182	32	3.7	0.3507	0.7150	11.6732
3	7.3	0.0000	1.4164	14.2371	33	2.6	0.0000	0.3300	14.9833
4	3.6	0.3319	0.7379	10.7645	34	4.7	0.0000	0.3300	13.0373
5	2.4	0.1462	0.5011	10.2909	35	4.2	0.2184	0.7060	12.4846
6	6.2	0.0000	0.5683	14.1763	36	2.1	0.6744	0.4116	10.0751
7	3.9	0.1240	1.0400	11.2901	37	4.5	0.6990	0.6382	12.6159
8	3.5	0.4734	0.6862	11.5486	38	1.4	0.0525	0.4606	9.8261
9	3.4	0.5154	0.3300	11.5286	39	3.2	0.0000	1.0734	10.5814
10	4.3	0.0000	0.6605	11.9604	40	3	0.2582	0.6760	11.5741
11	5.2	0.0000	0.5125	13.7166	41	3.9	0.0495	0.7025	11.5950
12	5.1	0.0000	0.5463	13.5197	42	1.3	0.1035	0.5093	10.5170
13	7.4	0.2307	0.9863	14.8000	43	3.3	0.3872	0.4222	11.2374
14	3.1	0.2624	0.6977	13.1720	44	1.8	0.1423	0.5384	12.2803
15	7.2	0.9682	1.7351	14.8000	45	4.6	0.3748	0.3300	13.3413
16	4.8	0.0000	0.8545	12.3239	46	3.6	0.2935	0.6949	10.6030
17	3.6	0.6152	0.3300	12.0980	47	5.2	0.0134	0.6500	13.7419
18	4	0.0705	0.5799	12.4204	48	6.7	0.8321	0.7890	12.3442
19	5.9	0.0777	0.5023	13.9459	49	7.3	0.0721	1.2715	14.5382
20	3	0.0554	0.5144	11.4723	50	7.8	0.0000	1.5171	14.8000
21	3.5	0.1243	0.7088	11.3351	51	4.4	0.0305	0.6726	12.8877
22	3.3	0.0000	0.7437	11.1333	52	3	0.5124	0.5925	10.6401
23	2.5	0.9780	0.3300	13.2608	53	5.9	0.2403	1.2058	12.9878
24	4.4	0.0122	0.7465	11.9858	54	2.9	0.0000	0.3726	11.8342
25	6	0.0092	0.6802	14.2501	55	2.2	0.0000	0.5705	10.6047
26	3	0.3952	0.7722	10.9762	56	3.3	0.2287	0.5861	12.4923
27	5.1	0.3819	0.7281	13.3636	57	3.2	0.0000	1.0631	13.5835
28	4.4	0.5060	0.6421	12.2581	58	2.8	0.3629	0.5208	10.4389
29	2.5	0.0000	0.5677	11.2525	59	3.2	0.6648	0.5436	11.4785
30	2.7	0.0000	0.6478	11.4967	60	5.4	0.0948	0.8911	13.4889

Appendix A

Marking Guide:

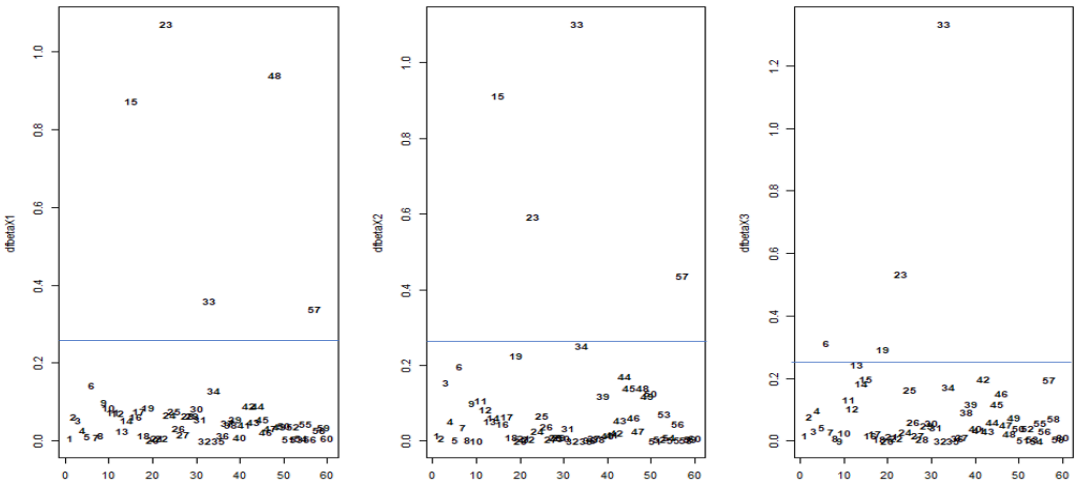
(a) An influential observation is an observation for a statistical calculation whose deletion from the dataset would noticeably change the result of the calculation. In particular, in regression analysis an influential point is one whose deletion has a large effect on the parameter estimates.

(b) Perform an analysis of influence on the individual coefficients, showing your working and justification clearly. Identify potential outliers.

$$DFBETAS_{j,i} = \frac{\hat{\beta}_j - \hat{\beta}_{j,(i)}}{\sqrt{s_{(i)}^2 C_{jj}}}$$

A large value in DFBETAS indicates a significant difference in estimated value when i-th observation is excluded (i.e. the i-th observation is influential on the j-th coefficient). $|DFBETAS_{j,i}| > 2/\sqrt{n}$ is the suggested cutoff.

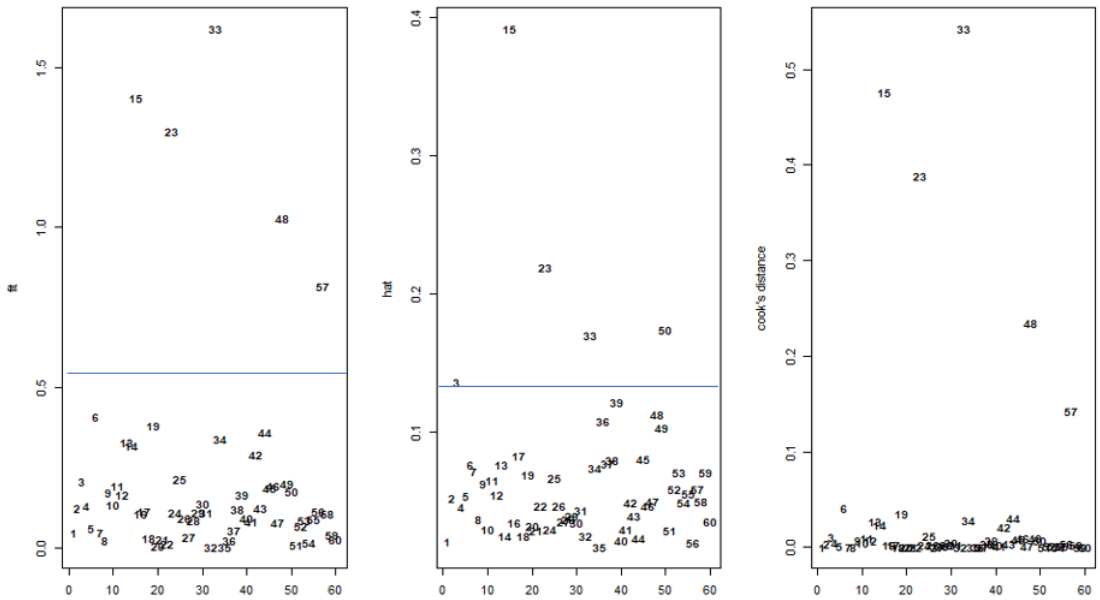
$$\frac{2}{\sqrt{n}} = 0.25819$$



Points 33, 38, 15, 23, 57 will result in a large difference in $\hat{\beta}_1$ when removed.
Points 33, 15, 23, 57 will result in a large difference in $\hat{\beta}_2$ when removed.
Points 33, 23, 6, 19 will result in a large difference in $\hat{\beta}_3$ when removed.

Points which consistently appear on all betas (33, 23) are likely to have either outlier values for all covariates or it might be just the outcome variable. Points unique to 1 beta (6, 19) might have a single outlier value for that particular covariate.

(c) Perform an analysis of influence on the model as a whole, showing your working and justification clearly. Identify potential outliers.
Extreme DFFITS > 2 sqrt(p/n) = 0.5164
Extreme Cook's D = qf(0.5,4,55) = 0.8496



Under DFFITS, samples 33, 15, 23, 48, 57 might be outliers
None of the samples have high enough cook's distance.

(d) From your analysis in part b and c, you will have identified potential outliers. Also attached in this question is the raw dataset which was used by the analyst. By referring to the raw dataset, discuss on the outliers that you have observed.

Potential outliers:

N	Y	X1	X2	X3	influence	Comments
15	7.2	0.9682	1.7351	14.8000	All	Very high in everything. X2 1.73 seems a little too high. Maybe typographical error?
23	2.5	0.9780	0.3300	13.2608	All	Inconsistent: High X1 and X3, low Y and X2.
33	2.6	0.0000	0.3300	14.9833	All	Inconsistent: Low Y, X1, X2 but highest X3
38	1.4	0.0525	0.4606	9.8261	beta1	Low in everything. X3 might be a little too low?
48	6.7	0.8321	0.7890	12.3442	All	Data looks ok
57	3.2	0.0000	1.0631	13.5835	All	Inconsistent: Low Y, X2 is slightly high ($\hat{\beta}_2$ is the largest coef)
6	6.2	0.0000	0.5683	14.1763	beta3	Inconsistent: High Y, X2 is slightly low ($\hat{\beta}_2$ is the largest coef)
19	5.9	0.0777	0.5023	13.9459	beta3	Same as 6.

Overall, 48, 57, 6, 19 appears to be within acceptable range. 38 and 15 are the extremes in the dataset, which we should take a closer look at the data source. 23 and 33 have low wine scores despite high X3.

To markers: There is no right answers here, only wrong answers.

(e) Some of the inconsistencies might actually be due to lack of data. For example, 23 and 33 have low scores – which should be driven by some important factor not captured in the data. They might not be outliers once we capture these information.

We should definitely take a closer look at the extreme values 38 and 15, however, their numbers are still within the "possible" region, so it might just be that 1 in 100 chance.

We should also attempt using other models, such as those with robust distributions, which might decrease the impact of outliers.

----- END -----